Multimodal Stock Forecasting with Financial News and Price Sequences: A Controlled Comparison of Fusion Architectures

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Abstract

Accurate stock forecasting depends on heterogeneous signals that range from raw market histories to unstructured financial news and structured corporate disclosures. This paper presents a standalone study of multimodal financial forecasting using the FinMultiTime benchmark. We evaluate price-only baselines, a text-only sentiment model, zero-shot time-series foundation models, and three strong multimodal price–news fusion architectures under a fixed chronological protocol on S&P 500 equities. To make the modeling contribution clearer, we formalize the forecasting task, describe the common news-encoding pipeline, and explain the design trade-offs behind TCN-FinBERT, TFT-FinBERT, and FinBERT+CNN+Transformer in greater detail. Across all evaluated systems, TCN-FinBERT achieves the lowest RMSE of 0.03029, followed by FinBERT+CNN+Transformer at 0.0316, while weaker late fusion and zero-shot foundation-model transfer lag behind. The results support a consistent conclusion: in multimodal financial forecasting, the mechanism used to align price dynamics with textual information matters at least as much as the temporal backbone itself. This paper therefore offers both an empirical benchmark and a practical design guide for future multimodal forecasting systems in finance.

1 Introduction

Financial time-series forecasting remains difficult because asset prices respond to multiple information channels simultaneously, including recent price dynamics, breaking news, investor sentiment, and firm-level fundamentals. Price-only models can capture local temporal regularities, but they often miss abrupt sentiment shifts and information shocks that arrive through text or disclosures [13, 15]. Recent multimodal benchmarks such as FinMultiTime make it possible to study these channels jointly by aligning time series, text, and structured financial data at scale [16].

Despite this progress, it is still unclear whether performance gains come primarily from richer data, from stronger model classes, or from the way modalities are fused. Many multimodal studies introduce a new architecture and a new data recipe at the same time, which makes attribution difficult [5, 11]. This study addresses that issue through a controlled benchmark on S&P 500 stocks: the prediction target, temporal split, and evaluation pipeline are held fixed while different unimodal and multimodal architectures are compared.

This paper makes three contributions. First, it provides a clear mathematical task definition for multimodal short-horizon stock forecasting. Second, it explains the three strongest multimodal models in more depth, including how each model encodes price histories, aggregates news, and maps fused representations to a five-day forecast. Third, it shows that the mechanism used to inject news into a temporal forecaster is the dominant design choice: structured fusion with FinBERT consistently improves over strong price-only baselines, whereas zero-shot foundation models and weak late fusion lag behind.

2 Problem Formulation

Let i index a stock and t index the last day in a lookback window. Each supervised example is a tuple

$$\mathcal{S}_{i,t} = \big(\mathbf{P}_{i,t},\, \mathcal{N}_{i,t},\, \mathbf{m}_i,\, \mathbf{y}_{i,t}\big),$$

where $\mathbf{P}_{i,t} \in \mathbb{R}^{T \times d_p}$ denotes the historical price-feature matrix, $\mathcal{N}_{i,t}$ denotes the aligned set of news articles in the window, $\mathbf{m}_i$ collects static metadata such as ticker and sector identity, and

$\mathbf{y}_{i,t} \in \mathbb{R}^H$ is the vector of future close prices. In this study, $H = 5$ trading days and $T \in \{30, 60, 90\}$ depending on the model family.

The forecasting function is written as

$$\hat{\mathbf{y}}_{i,t} = f_\theta(\mathbf{P}_{i,t}, \mathcal{N}_{i,t}, \mathbf{m}_i),$$

where f_θ may reduce to a price-only model or a text-only model when some inputs are omitted. Several architectures in this study forecast residual increments rather than prices directly. In that case, the model outputs $\Delta_{i,t,1:H}$ and reconstructs future closes as

$$\hat{x}_{i,t+h} = x_{i,t} + \sum_{u=1}^{h} \Delta_{i,t,u}, \qquad h = 1, \ldots, H.$$

This direct multi-step formulation avoids the error compounding that would arise if the next prediction were recursively fed back as input.

The primary evaluation metric is root mean squared error,

$$\mathrm{RMSE} = \sqrt{\frac{1}{NH} \sum_{i=1}^{N} \sum_{h=1}^{H} (y_{i,h} - \hat{y}_{i,h})^2},$$

reported together with MSE and MAE in our evaluation. We emphasize RMSE in this paper because it captures the overall model ranking most clearly across the benchmark.

3 Related Work

Classical and neural stock forecasting methods have long relied on historical market features. Recurrent architectures such as LSTMs remain important baselines for financial prediction because they can model sequential dependence without requiring extremely large datasets [6, 4]. More recent sequence models have shifted toward patch-based Transformers and parameter-efficient mixing architectures, including PatchTST and TSMixer, which improve long-horizon forecasting in many generic time-series settings [12, 3]. Temporal Fusion Transformers (TFTs) extend this line of work with variable selection, gating, and interpretable attention for multi-horizon prediction [9].

On the text side, domain-specific language models such as FinBERT provide representations of financial sentiment and narrative context that are better aligned with market language than generic BERT-style encoders [7]. Recent multimodal stock-prediction systems therefore combine textual encoders with temporal backbones, often through attention, gating, or graph-based fusion [5, 11]. The main open question is not whether text helps at all, but how to inject it into a temporal forecaster without allowing the high-dimensional text embedding to dominate the price signal.

Table 1: FinMultiTime data summary used to define the project setting.

Dataset type	Size	Tickers	Frequency
Semi-annual reports	84.04 GB	2,676	Semi-annual
Stock time series	1.83 GB	4,213	Daily
Daily text	14.10 GB	4,694	Daily
Total	108.64 GB	—	—

Time-series foundation models such as Chronos and Chronos-2 have raised the possibility of zero-shot or lightly adapted forecasting across domains [2, 1]. However, evidence on financial equities remains mixed, with domain mismatch often leading to underperformance relative to supervised in-domain baselines [14]. This setting is therefore especially useful because it compares conventional supervised models, domain-specific language models, and foundation-model variants under one evaluation protocol.

4 Data and Preprocessing

The experiments are based on the FinMultiTime dataset [16]. The original benchmark spans multiple modalities and markets from 2000 to 2025, while this study restricts attention to S&P 500 equities and excludes the image modality in order to keep the comparison centered on price histories, news, and structured reports. All experiments use a chronological 70%/15%/15% train/validation/test split to avoid look-ahead bias.

Price inputs include OHLCV variables and technical indicators such as RSI, MACD, and Bollinger Bands. The study uses normalized per-stock features to improve stationarity and comparability across assets. News items are aligned to daily timestamps and either summarized into scalar sentiment labels or encoded directly by FinBERT. For the multimodal models emphasized here, the richer FinBERT embeddings are the main representation, because they preserve polarity, intensity, and contextual nuance better than a coarse ordinal label.

The preprocessing pipeline also constructs ticker and GICS-sector metadata. These variables are important because they provide static contextual information that price and news alone do not capture. Structured SEC-derived indicators were prepared during preprocessing, but they were not fully integrated into the three strongest quantitative models; we therefore treat them as an important direction for future work

rather than a core experimental variable in this paper.

5 Modeling Methods

We study a broad family of models including LSTM, PatchTST, TSMixer, Chronos, FinBERT, and several multimodal hybrids. For a concise conference-style presentation, we focus on the three strongest multimodal systems in the benchmark. All three share the same five-day prediction horizon and rely on FinBERT-derived news representations, but they differ in how price and text are combined and where inductive bias is placed.

5.1 Common News Encoding Pipeline

For any article n_j in the aligned news set, FinBERT produces a contextualized embedding $\mathbf{h}_j \in \mathbb{R}^{768}$. When multiple articles are available, the model pools them with learned attention rather than a plain average. A generic form of this aggregation is

$$\alpha_j = \mathrm{softmax}_j\left(\mathbf{w}^\top \tanh(\mathbf{W}\mathbf{h}_j)\right),$$
$$\mathbf{z}_{\text{news}} = \sum_j \alpha_j \mathbf{h}_j.$$

This design allows the system to focus on the most informative article within a dense news window, which is especially important in finance because many articles are repetitive or only weakly relevant to next-week price movement.

5.2 TCN-FinBERT

TCN-FinBERT is the best-performing model in this study. Each article is encoded with frozen FinBERT, and a four-head cross-article attention layer pools the article-level `[CLS]` vectors into a 768-dimensional news context. In parallel, engineered price features are processed by a hybrid temporal backbone composed of four dilated temporal convolution blocks followed by a two-layer Transformer encoder. The dilated convolutions capture short- and medium-range motifs such as reversals, breakouts, and volatility bursts, while the Transformer layers provide a lightweight mechanism for modeling longer-range interactions across the 60-day lookback window.

The final temporal state is pooled into a 256-dimensional context vector and concatenated with ticker and sector embeddings. A LayerNorm–MLP bottleneck maps the fused representation to a 5-step forecast trained with Huber loss [8]. Formally, if $\mathbf{z}_{\text{time}}$ is the pooled temporal representation, the fused vector takes the

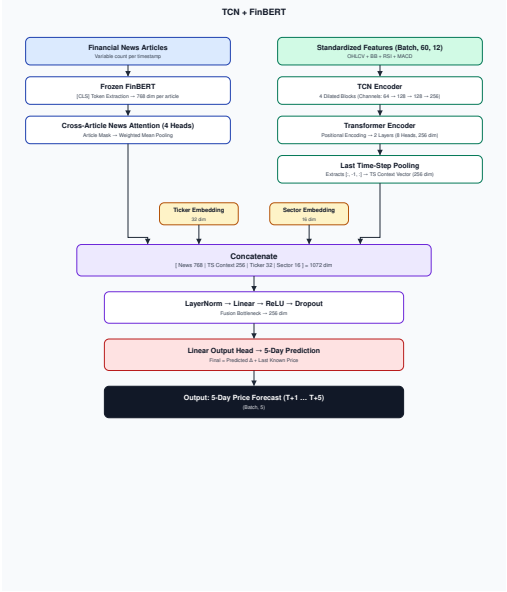

Figure 1: TCN-FinBERT architecture.

form

$$\mathbf{z}_{\text{fuse}} = [\mathbf{z}_{\text{news}}; \mathbf{z}_{\text{time}}; \mathbf{e}_{\text{ticker}}; \mathbf{e}_{\text{sector}}],$$

followed by a projection to a lower-dimensional bottleneck and a linear output head. The architecture is effective because it balances locality, global context, and a simple but stable fusion layer.

5.3 TFT-FinBERT

TFT-FinBERT replaces the TCN-Transformer branch with a Temporal Fusion Transformer backbone. The temporal path uses a Variable Selection Network, an LSTM encoder, gated skip connections, positional encoding, interpretable attention, and an output GRN [9]. This gives the model a strong bias toward feature selection and interpretability: the VSN can suppress irrelevant covariates, while the attention layer identifies the most important timesteps for the forecast.

The news branch again uses frozen FinBERT followed by cross-article attention pooling. The resulting news context, temporal context, ticker embedding, and sector embedding are concatenated, normalized, and projected to a 256-dimensional fusion bottleneck before a linear head predicts five future residuals. Relative to TCN-FinBERT, this architecture allocates more modeling capacity to the temporal encoder itself and less to explicit cross-modal interaction. That design is attractive for interpretability, but it may also explain why TFT-FinBERT trails the best model: the text signal is injected late, after the temporal representation has already been compressed.

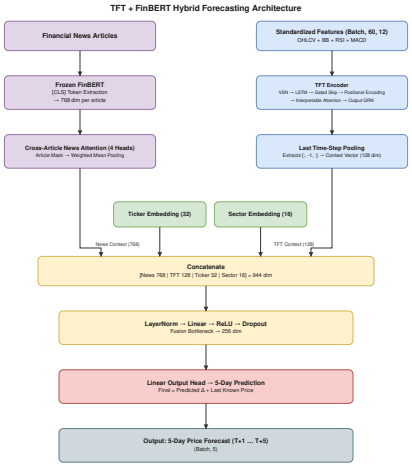

Figure 2: TFT-FinBERT late-fusion architecture.

5.4 FinBERT + CNN Price Encoder + Transformer Decoder

The FinBERT+CNN+Transformer model uses a richer encoder–decoder design. On the news side, daily articles are independently encoded by FinBERT and aggregated through learnable article attention within each trading day; positional encodings then create a time-indexed sequence of daily news embeddings. On the price side, the close-price history is passed through a one-dimensional CNN with convolution, batch normalization, ReLU, and global average pooling to form a compact price representation. The CNN branch is intentionally shallow, emphasizing local price-shape extraction rather than deep temporal compression.

Both streams are projected into a common latent space and summed to form the encoder memory. A Transformer decoder with $H = 5$ learnable query tokens attends to this memory and outputs residual price increments that are accumulated onto the last observed close. If $\mathbf{M}$ denotes the fused encoder memory and $\mathbf{q}_h$ denotes the query for horizon h, each step prediction is obtained through cross-attention followed by a scalar projection. This model is therefore the most explicit cross-modal interaction design in the benchmark: price and news are aligned before the horizon-specific decoder generates outputs.

5.5 Training Objective and Design Comparison

Most high-performing models in this study optimize a robust Huber objective rather than plain squared error. For a target y and prediction $\hat{y}$,

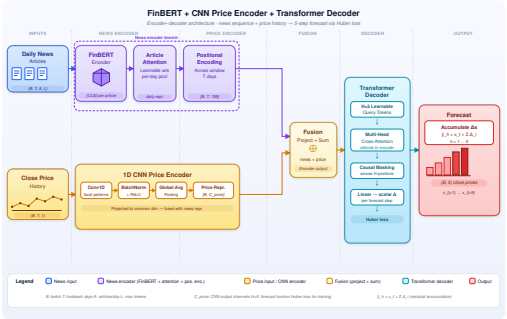

Figure 3: FinBERT+CNN+Transformer Decoder architecture.

Table 2: Comparison of the three highlighted multimodal models.

Model	Price encoder	Fusion strategy	Main strength
TCN-FinBERT	Dilated TCN + Transformer	Late fusion after strong temporal encoding	Best overall accuracy with balanced complexity
TFT-FinBERT	TFT with variable selection and attention	Late fusion after interpretable temporal compression	Interpretability and feature selection
FinBERT+CNN+Transformer	1D CNN price encoder + Transformer decoder	Encoder–decoder fusion with horizon queries	Richest cross-modal interaction across forecast steps

the scalar Huber loss is

$$\mathcal{L}_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & |y - \hat{y}| \le \delta, \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2, & \text{otherwise.} \end{cases}$$

This is appealing in equities because extreme market moves are common and can destabilize pure MSE training. In practical terms, the three main multimodal models differ less in objective than in where the fusion occurs and how much temporal structure is preserved before fusion.

6 Experimental Setup

All models are trained or evaluated on the same chronological split and forecast the next five trading days. Hyperparameters are tuned independently for each architecture using validation RMSE, and the best validation configuration is carried into test evaluation. This choice should be interpreted as a best-per-model comparison rather than a shared-grid ablation. The model zoo includes price-only baselines (LSTM, PatchTST, TSMixer, TFT, and TCN-Transformer), a text-only FinBERT baseline, zero-shot Chronos variants, a partially adapted Chronos-FinBERT hybrid, and the three multimodal models highlighted above.

A useful design feature shared by many baselines is direct multi-step forecasting. Instead of rolling out predictions autoregressively, the

Table 3: Reported evaluation metrics across the model families studied here. Lower is better for RMSE, MSE, and MAE.

Family	Model	RMSE	MSE	MAE
Price only	LSTM	0.03794	0.00143	0.02939
Price only	PatchTST	0.233	0.0543	0.1466
Price only	TSMixer	0.2228	0.0496	0.1474
Price only	Chronos-2	1.3505	1.8241	1.1954
Price only	Chronos T5-Large	1.3669	1.8684	1.2130
Price only	TFT	0.0681	0.00463	0.0358
Price only	TCN-Transformer	0.0673	0.00453	0.0355
Text only	FinBERT	0.1821	0.0332	0.1602
Price + news	Chronos-FinBERT	0.1249	0.0156	0.0418
Price + news	**TCN-FinBERT**	**0.03029**	**0.00091**	**0.0286**
Price + news	TFT-FinBERT	0.038212	0.00146	0.03146
Price + news	FinBERT+CNN+Transformer	0.0316	0.00099	0.0302

models predict the full five-day horizon in one pass. This makes the comparison cleaner, because differences in error do not simply reflect different rollout instability. Optimization primarily uses AdamW [10] with model-specific learning rates and manual tuning over architecturally motivated ranges.

A second experimental choice worth emphasizing is that each model is tuned independently on the validation split rather than under one global hyperparameter grid. This mirrors how practitioners actually deploy forecasting systems, but it also means that the benchmark should be interpreted as a comparison of best-found implementations rather than a perfectly controlled architecture-only study. That caveat is especially relevant when comparing the two best multimodal models: the small gap between TCN-FinBERT and FinBERT+CNN+Transformer is meaningful, yet both should be viewed as strong design points rather than as mutually exclusive winners under every tuning budget or market regime.

7 Results and Discussion

7.1 Quantitative Comparison

Table 3 summarizes the quantitative metrics used in this study. We report RMSE, MSE, and MAE for each evaluated model with available outputs. Per-model directional-accuracy values are not included because consistent numeric values are unavailable across the full benchmark. TCN-FinBERT still emerges as the best overall system by RMSE, followed closely by FinBERT+CNN+Transformer. Relative to its unimodal TCN-Transformer counterpart, TCN-FinBERT improves RMSE by roughly 55%, and it also outperforms the strongest price-only baseline (LSTM, 0.03794) by about 20%.

TFT-FinBERT remains competitive at 0.03821 and improves substantially over unimodal TFT, yet it does not match the best cross-modal designs. FinBERT on its own

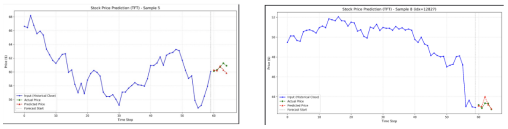

Figure 4: Representative TFT forecast examples.

reaches 0.1821 RMSE and 0.1602 MAE, which is worse than the best temporal models but still informative: news is predictive in isolation, just not sufficient for high-quality short-horizon forecasting. By contrast, zero-shot Chronos-2 and Chronos T5-Large perform poorly across all reported error metrics, which is consistent with the domain-mismatch explanation advanced in prior work on financial foundation models [14].

7.2 What the Three Main Models Teach Us

The three highlighted multimodal models provide a useful spectrum of design choices. TCN-FinBERT demonstrates that a relatively compact temporal encoder can become state of the art when paired with strong domain-specific text embeddings and a stable fusion bottleneck. Its advantage likely comes from the complementarity between dilated convolutions, which are good at capturing local market motifs, and a lightweight Transformer stage, which expands the receptive field without making optimization brittle.

FinBERT+CNN+Transformer comes very close to the top result despite using a much shallower price encoder. Its strength is not raw temporal depth but horizon-aware decoding: because the Transformer decoder receives dedicated query tokens for each future day, it can condition the generation of each step on a shared cross-modal memory. This is a more expressive mechanism than static concatenation and explains why the model remains highly competitive even though its price branch is simpler than the TCN-based alternative.

TFT-FinBERT, in contrast, is the most interpretable of the three. Its variable selection and attention mechanisms provide a clearer view of what features and timesteps matter, which may be valuable in high-stakes financial settings. However, the same design also compresses the temporal stream before fusion, and the final concatenation step may not be flexible enough to exploit the full 768-dimensional news representation. In that sense, TFT-FinBERT is a strong and principled baseline, but not the strongest architecture when pure predictive accuracy is the objective.

The qualitative behavior is also consistent with the quantitative ranking. Figure 4 shows

representative TFT forecasts on held-out stocks: the model tracks broad direction well but still smooths local peaks and troughs. Similar smoothing appears in LSTM and patch-based baselines, while the Chronos models frequently collapse toward the mean forecast trajectory. Taken together, the evidence suggests that the main bottleneck is not simply temporal modeling capacity; rather, it is the ability to preserve complementary signals across modalities without allowing one representation to dominate the others.

7.3 Why Foundation Models Lag in This Benchmark

The weak results for zero-shot Chronos-2 and Chronos T5-Large are informative rather than incidental. These models are pre-trained across broad time-series corpora, where learning a robust general temporal prior is useful. Equity data, however, exhibits heavy-tailed returns, abrupt regime changes, and a characteristically low signal-to-noise ratio. A model that defaults toward the center of its prior distribution may look well calibrated on many industrial forecasting datasets, yet the same behavior becomes unhelpful for short-horizon stock prediction, where relatively small directional deviations matter. The mean-collapsing forecast patterns are therefore consistent with a structural domain-mismatch explanation rather than a simple implementation failure.

The partially adapted Chronos-FinBERT hybrid supports this interpretation. Once the temporal encoder is anchored to in-domain data and paired with a domain-specific text representation, performance improves substantially over zero-shot Chronos. Even so, it still does not match the best supervised multimodal systems. This suggests that foundation models may be most useful in finance when they are treated as priors or feature extractors rather than as complete off-the-shelf forecasters. For practitioners, the message is pragmatic: pretrained temporal models are valuable, but only when paired with adaptation, feature engineering, and a fusion mechanism that respects the peculiar statistics of financial markets.

7.4 Limitations and Future Directions

This study also highlights three practical lessons for multimodal financial modeling. First, fusion strategy dominates backbone selection: structured cross-modal interaction beats simple late fusion. Second, domain-specific language representations matter: FinBERT is consistently useful across multimodal systems. Third, strong feature engineering remains important even in the presence of pretrained encoders. A carefully designed LSTM baseline remains surprisingly competitive, showing that multimodal gains must be measured against strong price-only baselines rather than weak temporal models.

There are also important limitations. The benchmark is restricted to S&P 500 stocks, which are unusually liquid and richly covered by news. Structured SEC XBRL-derived features were preprocessed but not fully integrated into the final quantitative comparison, leaving an important modality underexplored. Finally, the per-model tuning strategy improves practical performance but complicates strict architectural attribution. Future work should therefore extend the benchmark with filing-aware multimodal fusion, regime-shift evaluation, and ranking-based financial metrics in addition to point-forecast error.

8 Conclusion

This paper presents a benchmark study of multimodal financial forecasting. Across a fixed evaluation protocol on S&P 500 equities, the study finds that the best performance comes from combining strong temporal encoders with domain-specific text representations through carefully designed fusion layers. TCN-FinBERT and FinBERT+CNN+Transformer emerge as the most effective systems, while TFT-FinBERT offers a competitive and interpretable alternative. The broader takeaway is simple: in multimodal stock forecasting, the decision of *how* to combine price and text matters more than the choice of temporal backbone in isolation. A useful next step is to integrate structured filing data with the same care currently given to temporal and textual alignment.

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